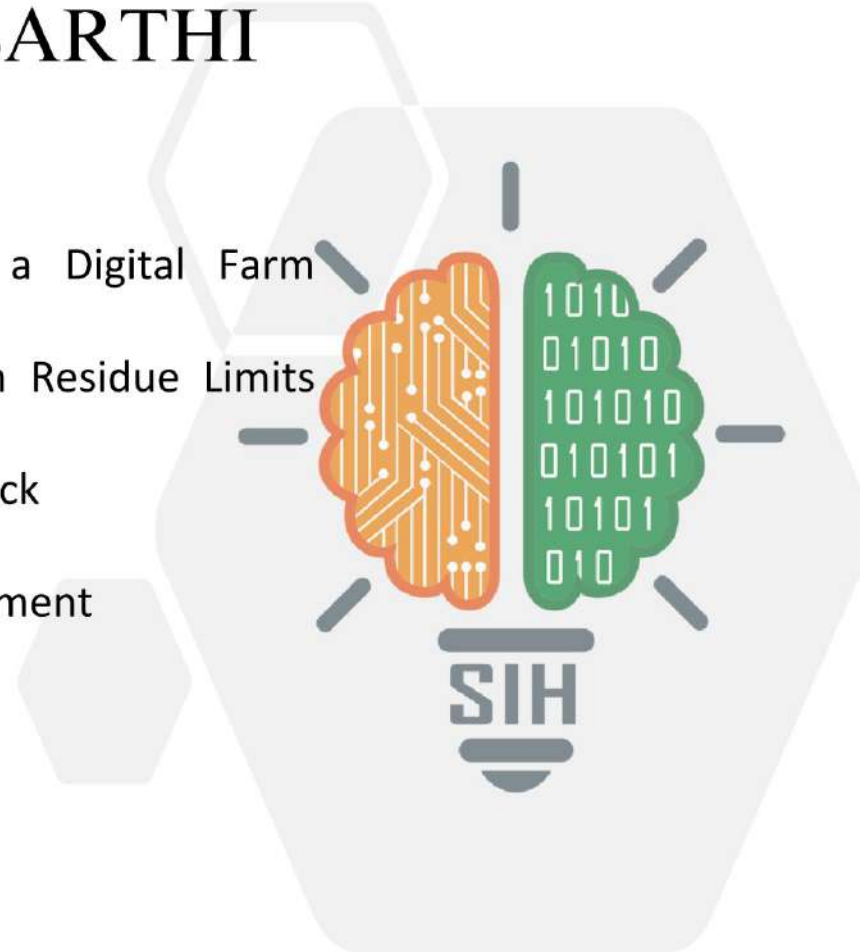


SMART INDIA HACKATHON 2025



JEEVSARTHI

- **Problem Statement ID –25007**
- **Problem Statement Title-** Development of a Digital Farm Management Portal for Monitoring Maximum Residue Limits (MRL) and Antimicrobial Usage (AMU) in Livestock
- **Theme-** Agriculture , FoodTech & Rural Development
- **PS Category-** Software
- **Team ID-93240**
- **Team Name-Cattle-Coders**





Cattle-Coders



JeevSarathi

"AI-Enabled, Multi-User Livestock Drug Monitoring & Compliance Portal"



JeevSarathi is an AI-enabled digital platform for livestock management that monitors Antimicrobial Usage (AMU) and Maximum Residue Limits (MRL) to ensure food safety. It connects farmers, veterinarians, and labs on a unified mobile-friendly portal, offering real-time compliance alerts. The platform empowers farmers with data-driven insights while promoting responsible livestock care and regulatory adherence.

PROPOSED SOLUTION

- Digital Portal & Mobile App
- User-Friendly designed for easy adoption by farmers.
- Real time Alerts when dosage or residue limits are breached.
- Unified Digital Ecosystem – Single plat- form for farmers ,vets,Labs and Administrators with secure role-based access.
- Automated Risk detection– Real-time calculation of MRL & AMU values , generating alerts on threshold breaches

MVPS

PASHUMITRA

- **Bridge Role:** PashuMitra connects farmers, veterinarians, and the digital portal at the village level.
- **Farmer Support:** Assists with prescriptions, safe drug usage, and digital record-keeping, especially for illiterate farmers.
- **Community Engagement:** Organizes vaccination camps with vets, spreads health awareness

SMALL FARMERS

- **Risk:** Farmers with 1–2 cattle face high income loss from MRL breaches.
- **Treatment:** Govt. hospitals provide temporary custody and care for affected cattle.
- **Support:** Farmers get some financial aid during the withdrawal period to protects farmer's livelihood.

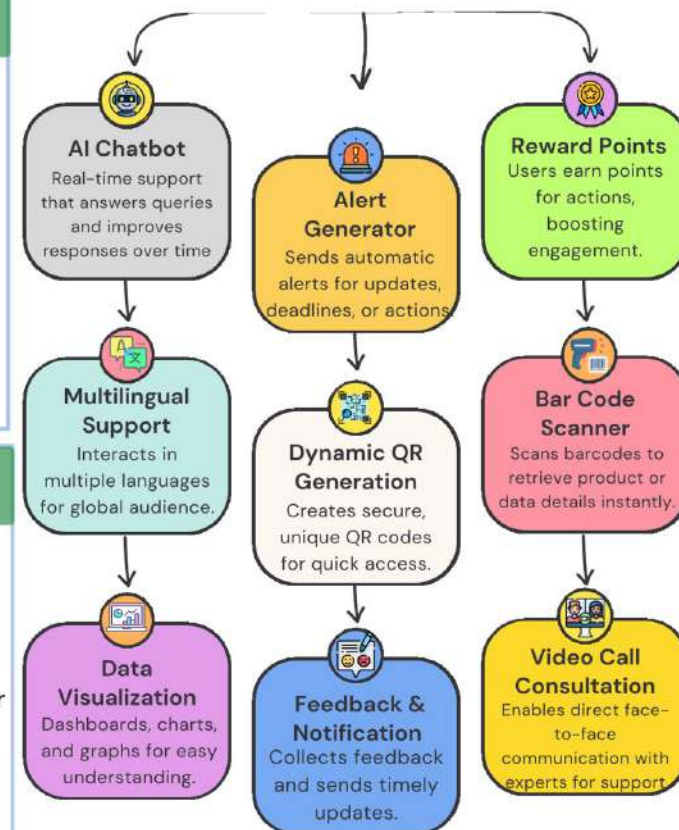
CAMPS

- **Pashu Jagruti Abhiyan:** Village-level health and awareness camps led by Pashu Mitra and veterinarians.
- **Health Services:** On-site mass vaccinations and checkups with real-time data entry.
- **Farmer Empowerment:** Training on drug safety, AMR, MRL, and hands-on digital literacy for using the mobile app.

PASHU SWASTHYA CARD

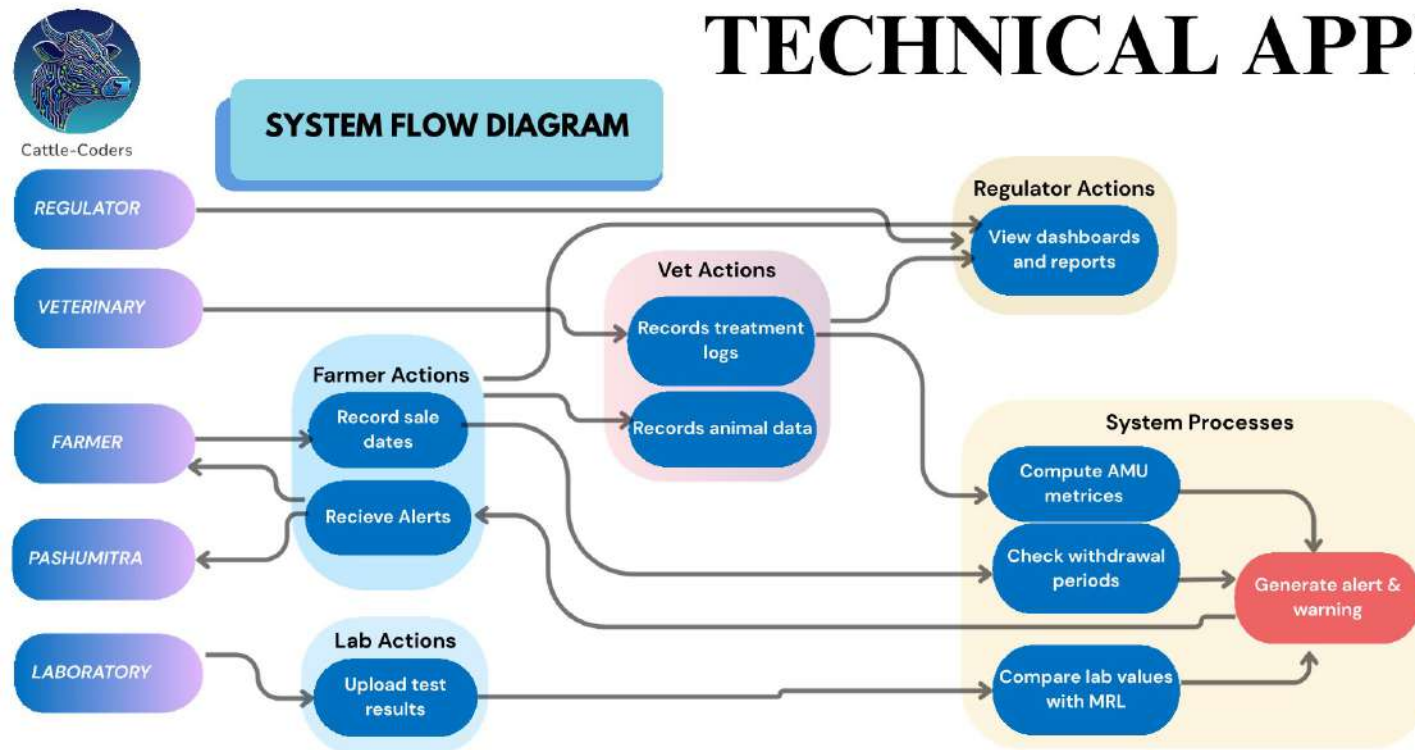
- **Unified Digital ID:** Pashu Swasthya Card links farmers to the national animal health portal.
- **Animal Registry:** Maintains a digital record of all animals via Pashu Aadhaar IDs.
- **Medicine Regulation:** Logs veterinary drug purchases and sets limits based on herd size.

Innovative ideas



TECHNICAL APPROACH

SYSTEM FLOW DIAGRAM

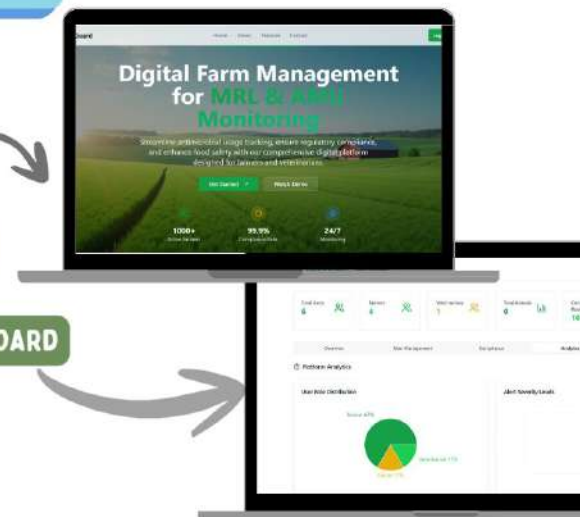


PROTOTYPE

<https://farm-safe-sync.vercel.app/>

LANDING PAGE

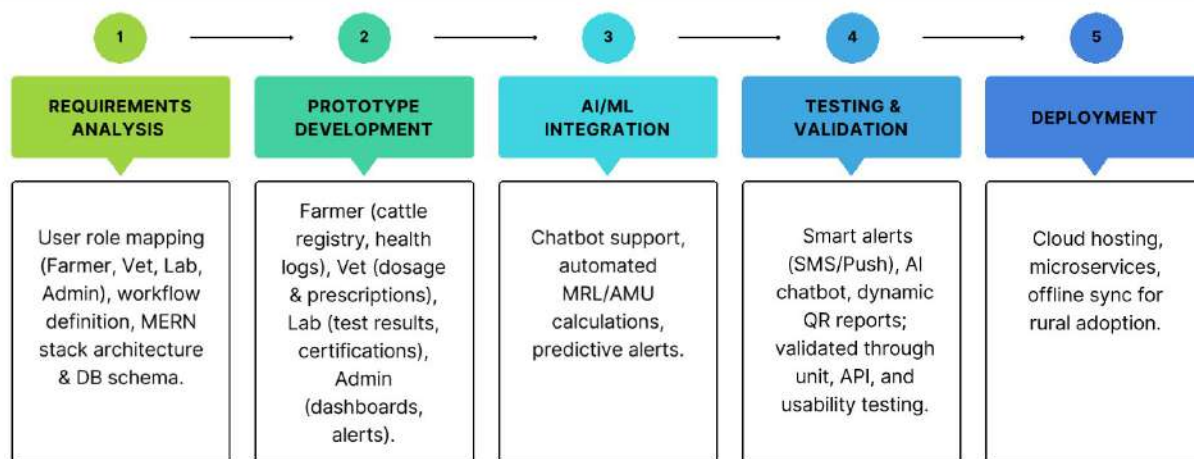
DASHBOARD



TECH STACK



Methodology & Process





FEASIBILITY AND VIABILITY



Feasibility Analysis

- 1 Video Consultation:** Farmers can video call vets for advice.
- 2 SMS Alerts :** Send important alerts via text message, no internet needed.
- 3 Modern Tech Stack:** Build a scalable, secure, and maintainable platform.
- 4 Accessible Design:** Ensure intuitive use for all levels of technical literacy.
- 5 Multi-language Support:** Accommodate a diverse, global user base.
- 6 Role-Based Access:** Provide customized workflows for different user types (vets, owners).
- 7 AI-Powered Support:** Offer 24/7 automated assistance via an integrated chatbot.



Potential Challenges

- 1 Farmer Adoption and Digital Literacy:** Overcoming resistance to new technology in traditional settings.
- 2 Data Integrity and Accuracy:** Ensuring the accuracy and reliability of user-entered data.
- 3 Internet in Rural areas:** Mitigating limitations of poor internet infrastructure.
- 4 Legacy System Integration:** Managing technical complexity when connecting to older software.
- 5 Model Training :**End-to-end ML model development executed on Vertex AI.



Viable Strategies

- 1 User Friendly:** Designed for simplicity and ease, making every interaction smooth
- 2 Color Coding :**Clear color coding helps users quickly interpret and understand data at a glance.
- 3 QR For Food Safety :** Enhance food safety and supply chain transparency.
- 4 Incentivization:** Motivate engagement and data accuracy through rewards.
- 5 Efficient Algorithm :** Supervised machine learning using linear regression
- 6 Data Monetization :** Generate revenue from aggregated, anonymized industry insights.
- 7 Scalable Architecture :**Build a future-proof system to support long-term growth.



FUTURE SCOPE

INTELLIGENT ALERT SYSTEM FOR POTENTIAL RISKS

An AI-driven alert system predicts MRL breach risks in real time, ensuring farmer safety and public health.

GEO-TAGGING OF SERVICES

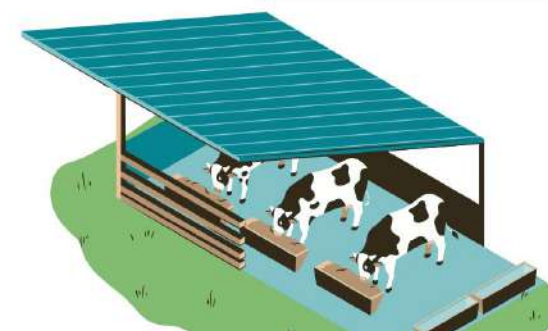
Map vaccination camps, vet visits, and outbreaks for better planning and resource allocation.

DIGITAL HEALTH CARDS FOR MILK & MEAT VIA QR CODE

Link animal health data to product safety certificates, boosting consumer trust and market value.

AUTOMATED DRUG ALERTS

Notify farmers when medicine doses or withdrawal periods are due.



IMPACTS AND BENEFITS

National and Economic impact

- Boosts exports by complying with EU/US safety standards.
- Unlocks new trade opportunities via blockchain-backed traceability.
- Strengthens livestock economy and rural livelihoods..
- Supports government priorities through digital compliance.
- Attracts foreign investment in agri-tech innovations.

Farmers

- Safeguards supply chain with residue-free milk, meat, and eggs.
- Mitigates antimicrobial resistance via systematic surveillance.
- Protects public health by preventing cross-species bacterial transfer.
- Enhances disease control through real-time alerts and early interventions.
- Ensures nationwide compliance with FSSAI 2025 standards

Technological & Innovation

- AI monitors anomalies for intelligent oversight.
- Blockchain provides tamper-proof treatment logs.
- Offline-first, mobile-friendly design enables rural adoption.
- Scalable microservices allow nationwide implementation.
- Future-ready with predictive analytics and forecasting.

Consumer

- Assured safe, residue-free dairy, meat & poultry.
- Builds trust in domestic markets via transparent traceability.
- Strengthens confidence in exports meeting global standards.
- Empowers consumers with certified, informed choices.

Environmental & Societal

- Reduces antibiotic residues in soil and water.
- Empowers rural communities while protecting livelihoods.
- Safeguards food sovereignty against contamination risks.
- Builds trust in governance through transparent monitoring.
- Advances One Health linking human, animal, environmental health.

Veterinarians

- Access to complete health history for accurate diagnosis.
- Digital formulary ensures guideline-based prescriptions.
- Auto withdrawal calculators minimize errors & risks.
- RFID "Scan & Treat" speeds up accurate treatment logging..
- Offline compliance assistant for remote farm visits.

Public Health & Food Safety

- Ensures residue-free milk, meat, and eggs.
- Curtails antimicrobial resistance via systematic monitoring.
- Enables early disease containment through real-time alerts.
- Guarantees nationwide compliance with FSSAI 2025 standards.

Regulators

- Proactive enforcement with real-time alerts.
- AMU dashboards provide regional compliance visibility.
- Heatmaps highlight high-risk misuse zones.
- Aggregated analytics support national policymaking.
- Vet verification ensures licensed prescribing only.





Cattle-Coders

RESEARCH AND REFERENCES



GAP ANALYSIS OF EXISTING DIGITAL PLATFORMS

Feature for MRL/AMU Monitoring	DAHD/State Websites	INAPH / Bharat Pashudhan	Proposed SIH 25007 Portal (JEEVSARTHI)
Unique Animal ID (Pashu Aadhaar)	✗ Not available	✓ Present	✓ Leverages existing Pashu Aadhaar
Field-Level Data Entry (Mobile)	✗ Not supported	✓ Available	✓ Supported (farmer-friendly mobile app)
Integrated Veterinary Drug Formulary	✗ Not available	✗ Not available	✓ Integrated formulary (all medicines & doses)
Dynamic Withdrawal Period Calculator	✗ Absent	✗ Absent	✓ Auto-calculation based on drug data
Real-Time Compliance Status	✗ Not available	✗ Not available	✓ Live compliance dashboard
Automated Farmer Compliance Alerts	✗ Not present	½ Only breeding alerts	✓ Withdrawal & safety alerts (multi-level)
Aggregated AMU Analytics Dashboard	✗ Not available	✗ Not available	✓ Advanced analytics dashboard



TEAM DETAILS :

•Team Leader:

Bhavya Kumar Singh

Branch: 3rd year CE

•Team Member:

Anurup Muwel

Branch: 3rd year CE

•Team Member:

Harsh Tamrakar

Branch: 3rd year CE

•Team Member:

Rachit Mahajan

Branch: 3rd year CE

•Team Member:

Chandrabhan Gadeshwar

Branch: 3rd year CE

•Team Member:

Saisha Deshpande

Branch: 1st year CE



RESOURCES:

•DAHD / State Websites:

dahd.nic.in

•INAPH / Bharat Pashudhan:

inaph.nddb.coop

•InFARM (FAO): fao.org

GLOBALG.A.P. RMS: globalgap.org

•FoodChain ID: foodchainid.com



PROJECT LINKS:

•Github Repo(All Resources):

: [github-repo-farm](https://github.com/repo-farm)

•Prototype (MVP Portal/App):

www.farmguard.com

•Live Demo(prototype model):

[youtube-link](#)



RESEARCH PAPER:

• [Antimicrobial Resistance in Livestock](#) :

A Serious Threat to Public Health

• [Global trends in antimicrobial use in food animals](#)

• [Blockchain technology in food safety and traceability concern to livestock products](#)

• [Blockchain-Based Frameworks for Food Traceability](#) : A Systematic Review



CLINIC VISIT:



To refine our project, we conducted primary research in Indore to understand real-world veterinary workflows. This fieldwork was crucial for developing a practical, user-centric solution.

- **Action Taken:** We visited multiple veterinary clinics and farms to observe daily operations and challenges.
- **Expert Consultations:** We engaged in detailed discussions with practicing veterinarians to gather firsthand insights.
- **Special Acknowledgments:** We are grateful for the invaluable expert guidance from **Dr. Rajanish Shrivastav** of Pet's Mate Clinic and **Dr. S.N. Banke**, whose feedback was instrumental.